From Mass-Spring Systems to Spectral Graph Neural Networks

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1 Mass-Spring System

1.1 The two particles system

Consider a spring follows Hook's law. Let two particles i and j connected by a spring located at x_i and x_j respectively and $e_{ij} = \frac{x_j - x_i}{||x_j - x_i||_2}$ be the direction from x_i to x_j then the force that i affects j can be represented as:

$$F_{ij} = -k(||x_j - x_i)||_2 - L)e_{ij} = -k(x_j - x_i) + kLe_{ij} \tag{1}$$

where k is a positive real number, the characteristic of the spring and L is the initial length the of spring. The magnitude of the force is proportional to the displacement from the initial distance between two particles.

Let two particles connected by a spring sit in an Euclidean space such that the particles can freely move on a particular z axis. At the initial condition, the two particles are located at x_i and x_j and the spring is at its length ($||x_j - x_i||_2 = L$) (no force).

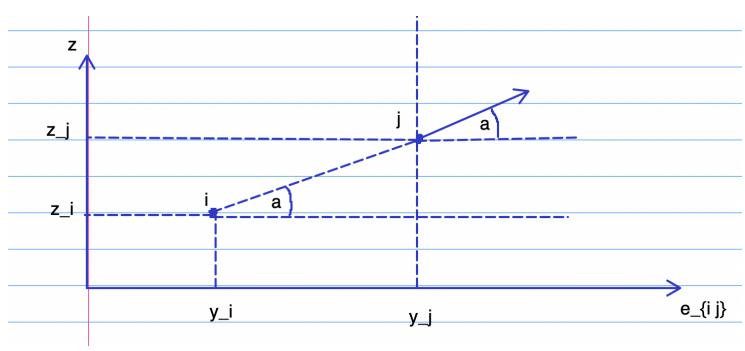

Figure 1: subspace of e_{ij} and z

Since, i and j can only move in the z axis, we can rewrite

$$x_i = x_i^{(0)} + z_i z$$
$$x_j = x_j^{(0)} + z_j z$$

Where $x_i^{(0)}$ and $x_j^{(0)}$ are the initial positions of i and j, z_i and z_j are the displacements on the z axis. Hence, the projected force on the z axis can be written as

$$F_{ij} \cdot z = (-k(x_j - x_i) + kLe_{ij}) \cdot z = (-k(x_j^{(0)} - x_i^{(0)}) + kLe_{ij}) \cdot z + (-k(z_j - z_i)z) \cdot z \tag{2}$$

The first term is the dot product of the initial force with the z direction which is essential zero since there is no force at the beginning. Hence, the projected force on the z axis can be written as

$$F_{ij} \cdot z = -k(z_j - z_i) \tag{3}$$

The projected force on the z axis linearly depends on the corresponding displacement.

1.2 The n particles system

Let n particles with the same weight m on an an Euclidean space that can freely move on a particular z axis. Some of them are connected by springs of the same characteristic k which is denoted by a undirected unweighted graph $G = (V, E)$. A particular node i is affected by all of its neighbours where the projected force on i is

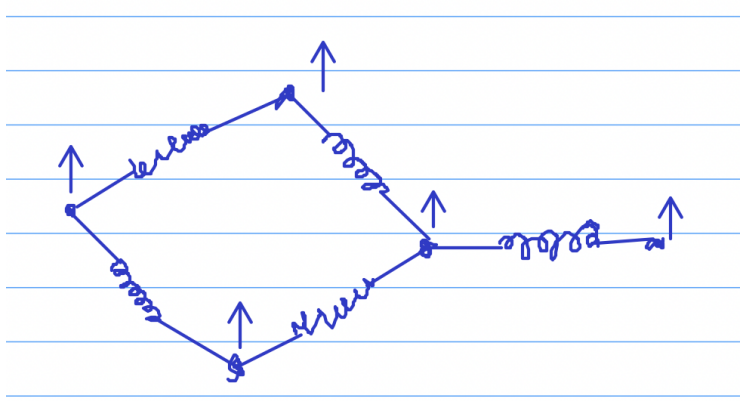

Figure 2: 4 particles system

$$F_i \cdot z = -k \sum_{e_{ji} \in E} z_i - z_j = -k(d_i z_i - \sum_{e_{ji} \in E} z_j) \tag{4}$$

Where d_i denotes the degree of the node i. Newton's Second Law of Motion:

$$a_i \cdot z = \frac{F_i}{m} \cdot z$$
$$\ddot{z_i} = -\frac{k}{m}(d_i z_i - \sum_{e_{ji} \in E} z_j)$$

We can rewrite in the matrix form

$$\ddot{z} = -\frac{k}{m}(D - A)z = -\frac{k}{m}Lz \tag{5}$$

Where $z = (z_1, z_2, ..., z_n)^T$, A is the adjacency matrix of G and D is the degree matrix of A (diagonal matrix where each entry equals to the corresponding degree of the node). $L = D - A$ is called Laplacian matrix of A.

We are seeking the mode of oscillation of the system. A mode of oscillation is a particular frequency where all particles oscillate at the same frequency. At that frequency, the differential equation for each particle must be in the form:

$$\ddot{z_i} = -\omega^2 z_i \tag{6}$$

Where ω is the frequency. In the matrix form:

$$\ddot{z} = -\omega^2 z \tag{7}$$

From 5 and 7, we have

$$Lz = \frac{m}{k}\omega^2 z \tag{8}$$

From 8, the oscillation mode frequencies are equivalent to the eigenvalues of the Laplacian matrix, and the initial condition to achieve each of the frequencies is the corresponding eigenvector.

Since L is real symmetric, by the Spectral Theorem, it has an eigenbasis. Furthermore, L is positive semi definite, then all of its eigenvalues are positive, hence the frequencies make sense.

For an arbitrary initial condition, since the system is linear, we can decompose the displacement z into the eigenbasis of the Laplacian matrix then solve each of the component individually.

2 Graph Laplacian Basis

Recall that, the eigen decomposition of L is as follow:

$$L = U\Lambda U^T = U\Lambda U^{-1} \tag{9}$$

Where each column vector in U is a normalized eigenvector.

Analogous to Fourier Transform, the eigenvalues of Laplacian matrix can serve as the frequency and the eigenbasis is corresponding to the Fourier basis.

Let $x \in \mathbb{R}^n$ be a graph signal on $G = (V, E)$ where each component of x is a real number corresponding to a node in G.

The convolution in spatial domain is equivalent to multiplication in spectral domain. Define the convolution operation as:

$$y(x) = U(U^T w \odot U^T x) \tag{10}$$

Where $w \in \mathbb{R}^n$ is called filter or kernel. Define $W = diag(U^T w)$ be the diagonal matrix whose entries are the entries of $U^T w$, we can rewrite 10 as

$$y(x) = (UWU^T)x \tag{11}$$

2.1 ChebNet

Let $\mathcal{L}$ be the normalized laplacian matrix.

$$\mathcal{L} = D^{-\frac{1}{2}} L D^{-\frac{1}{2}} = I - D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \tag{12}$$

The decomposition of $\mathcal{L}$

$$\mathcal{L} = U \Lambda U^T = U \Lambda U^{-1} \tag{13}$$

Theorem 1 *(Chung [**?**]) All eigenvalues of $\mathcal{L}$ are in the interval* $[0, 2]$.

ChebNet [**?**] approximate the diagonal matrix W using Chebyshev polynomials as the orthogonal basis in the polynomial subspace of the vector space of all functions $f : [-1, +1] \to \mathbb{R}$ with respect to the inner product.

$$\int_{-1}^{+1} f(x) g(x) \frac{dx}{\sqrt{1 - x^2}} \tag{14}$$

Chebyshev polynomials of the first kind:

$$T_{n+1}(x) = 2x T_n(x) - T_{n-1}(x) \tag{15}$$

Where $x \in [-1, +1]$, $T_0(x) = 1$ and $T_1(x) = x$. Let $f_i : [-1, +1] \to \mathbb{R}$ is an arbitrary function such that $f_i(\tilde{\lambda}_i) = w_i$ where $\tilde{\lambda}_i = \frac{2\lambda_i}{\lambda_{\max}} - 1 \in [-1, +1]$, $\lambda_{\max}$ is the largest eigenvalue. We want to project f_i into the subspace with the orthogonal basis of the first K terms of Chebyshev polynomials of the first kind. We can write f_i as

$$\hat{f}_i(t) = \sum_{k=0}^{K-1} \theta_{ki} T_k(t) \tag{16}$$

Hence, w_i is approximated as

$$\hat{w_i} = \hat{f}_i(\tilde{\lambda_i}) = \sum_{k=0}^{K-1} \theta_{ki} T_k(\tilde{\lambda_i}) \tag{17}$$

Matrix form of the approximation on W:

$$\hat{W} = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\Lambda}) \tag{18}$$

Where $\tilde{\Lambda}$ is the diagonal matrix of $\tilde{\lambda_i}$. Moreover,

$$U\hat{W}U^T = \sum_{k=0}^{K-1} \theta_k U T_k(\tilde{\Lambda}) U^T = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathcal{L}}) \tag{19}$$

Where $\tilde{\mathcal{L}} = \frac{2\mathcal{L}}{\lambda_{\max}} - I$. It is a great exercise to prove the Chebyshev recurrence for $\tilde{\mathcal{L}}$:

$$T_{n+1}(\tilde{\mathcal{L}}) = 2\tilde{\mathcal{L}} T_n(\tilde{\mathcal{L}}) - T_{n-1}(\tilde{\mathcal{L}}) \tag{20}$$

Finally, The convolution operation is

$$y(x) = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\mathcal{L}}) x \tag{21}$$

The construction of ChebNet avoids decomposing the matrix L as compare to 10.

2.2 Graph Convolutional Network

Similar to ChebNet, GCN [**?**] limits $K = 2$ and sets $\lambda_{\max} = 2$ hence $\tilde{\mathcal{L}} = \mathcal{L} - I$.

$$\begin{aligned} U\hat{W}U^T &= \theta_0 T_0(\tilde{\mathcal{L}}) + \theta_1 T_1(\tilde{\mathcal{L}}) \\ &= \theta_0 I + \theta_1 \tilde{\mathcal{L}} \\ &= \theta_0 I - \theta_1 D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \end{aligned}$$

The convolution operation is

$$y(x) = \theta_0 x - \theta_1 D^{-\frac{1}{2}} A D^{-\frac{1}{2}} x \tag{22}$$

The notes here is greatly inspired by [**?**].